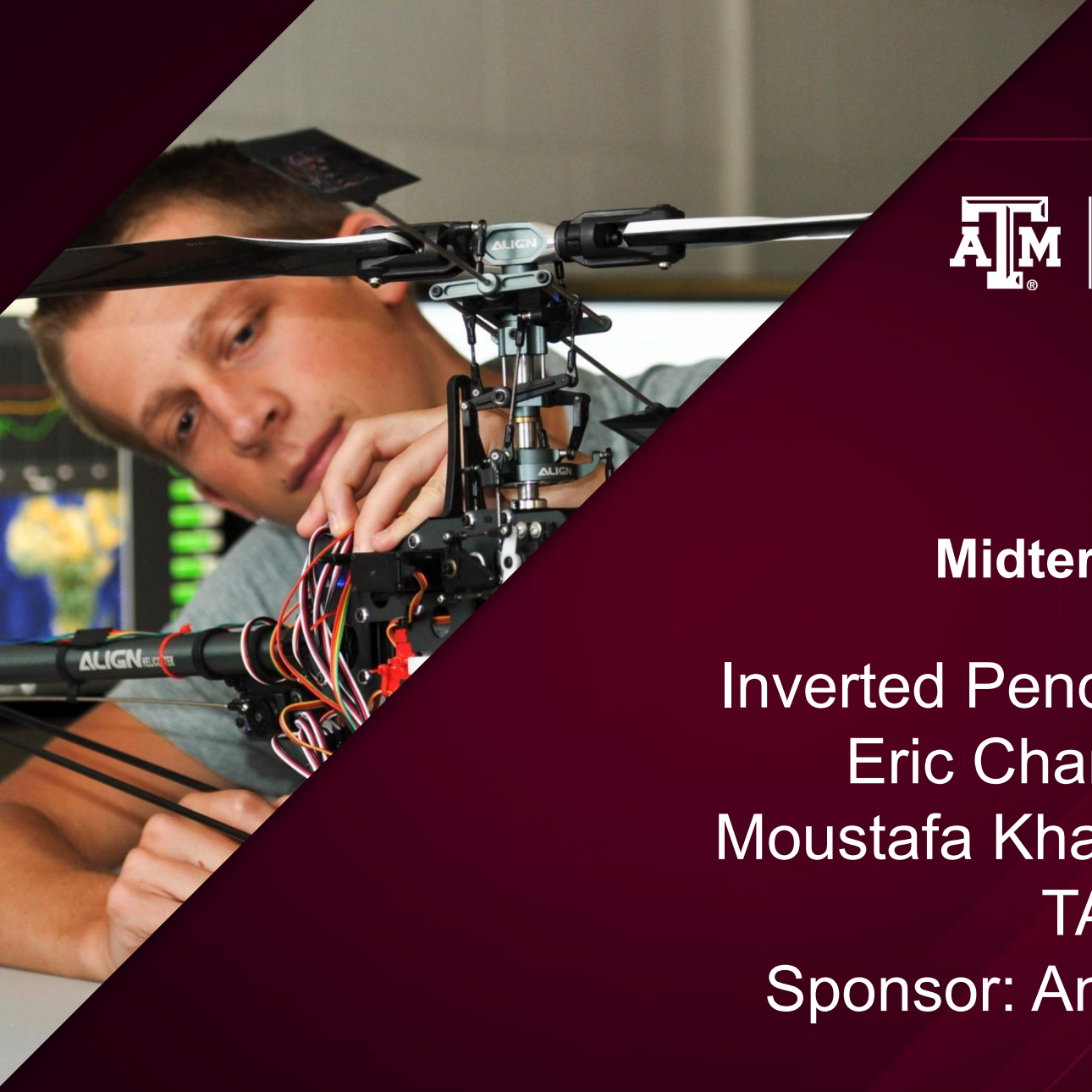

Dwight Look College of
ENGINEERING
TEXAS A&M UNIVERSITY

ECEN 403

Midterm Presentation

Inverted Pendulum Robot 1

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Sponsor: Aniruddha Datta

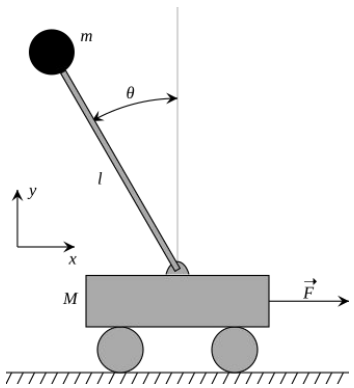
Project Summary

Problem Statement:

- How can a compact, mobile robot operate autonomously in tight, cluttered environments while keeping an inherently unstable load safe and upright
- Real settings (warehouses, labs, hospitals, inspection ducts) demand precise maneuvering, sudden stops/turns, and strict safety.

Our system will:

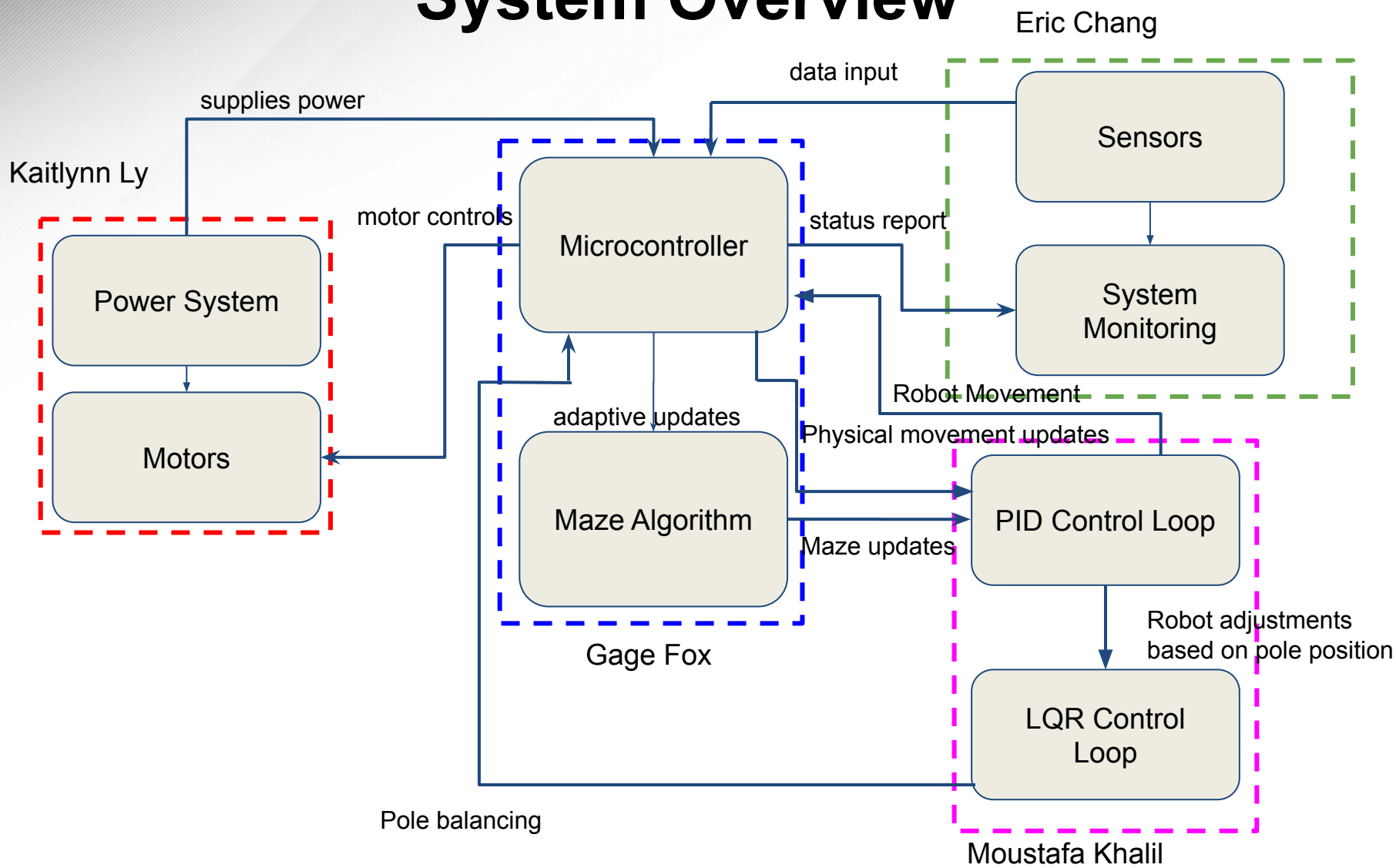
- Deliver an autonomous, self-navigating robot that maintains stability of an unstable payload while moving through constrained spaces.
- Integrate mechanical, sensing, power, and control subsystems into a cohesive, microcontroller-based platform designed for reliability and safety.



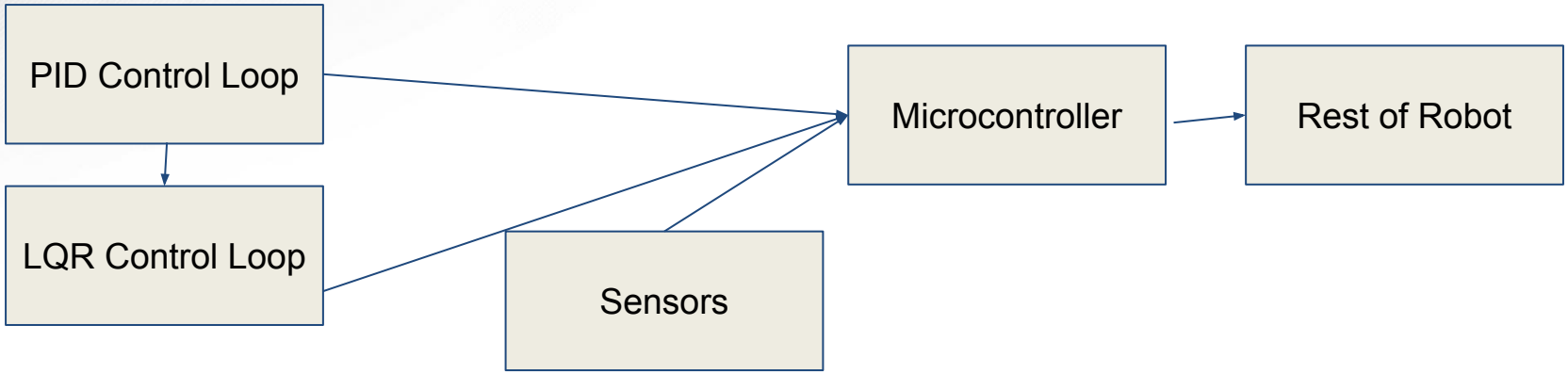
Side View

Isometric View

System Overview

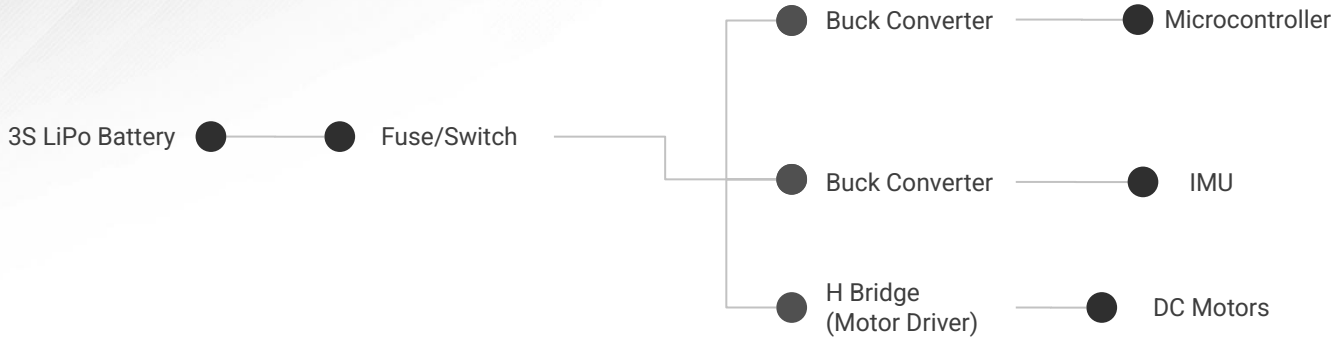


Controls – Moustafa Khalil



- This system will keep the 1m pole upright while commanding safe forward speed and turning so the robot can navigate the maze.
- It will have a two-loop controller with LQR/cascaded PID for the inner loop and a PID for the outer loop.It will implement C++ on a single MCU using the different sensors
- Maintain stable pole balance and smooth maze navigation
- The control algorithm ingests the data from the sensors and outputs a speed command plus safety flags to the MCU and power systems
- I chose the LQR over the PID for the inner loop for the better coupled balance and predictable turning. The PID requires more hand tuning and derivative filtering.

Power System & Mechanical - Kaitlynn Ly



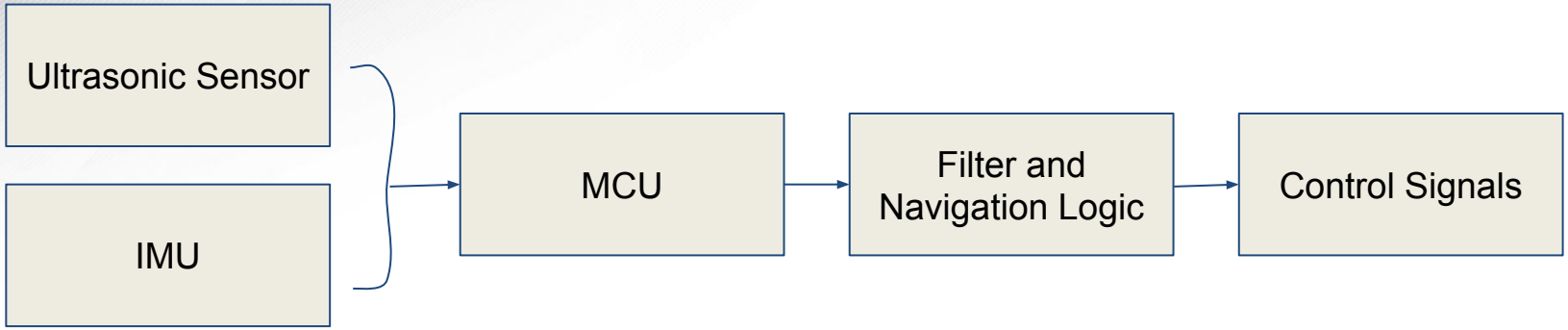
- Distributes battery power, provides structural frame to support pendulum/electronics, drives robot with DC motors
- Power: 3s LiPo battery with buck converters, fuse & switch, 4 DC motors, H bridge motor driver
  Mechanical: Two tiered aluminum disk with 3D printed smaller components, 4 wheels
- Keep robot powered and balanced within battery and weight limits by run time and motor strength
- Provides power to electronics, links motors to driver, and supports pendulum

Microcontroller – Gage Fox

Microcontroller Tradeoffs		
CHARACTERISTIC	YES	NO
Power Efficient	O	
High DMIPS	O	
High Frequency	O	
High IO Pin Count	O	
Precise	O	
Cheap		X
Small		X
Temp Resistant		X
High-Level Programming	O	
Large Voltage Range		X
Easy Implementation	O	

- Microcontroller: STM32F407ZGT6 (LQFP footprint)
- PCB Design Software: Altium
- Programming IDE: STM32CubeIDE
- Desired system: fast-reacting, power efficient, and precise.
- Microcontroller will connect to peripherals through PCB runs (longterm)
- For prototyping, perfboards will be sufficient.
- Will use built in ADC inputs and PWM outputs for sensor reading and motor driver control. An external oscillator will drive the clock.
- Low resistance to temperatures.

Sensors & Navigation – Eric Chang



- Supports navigation logic, obstacle detection/avoidance, and displays data through GUI
- Collects real-time data from ultrasonic sensors (emits high-frequency) and IMU.
  - IMU: 6 AXIS IMU INERTIAL MEASURE UNIT; LSM6DSV320XTR
  - Sensor: ULTRASONIC DISTANCE SENSOR - 3.3; SEN-28798
- Extended Kalman and Mahony filters for noise reduction and accurate estimations
- Matlab/Python for prototyping; C++ for implementation
- Calibrate for responsiveness, power monitoring, detection accuracy
  - Tilt ±2°, ≥95% obstacle detection at ≥2 cm distance, <100 ms latency, ≤3% power error

Project Execution & Validation

Completed initial design, selected appropriate components, highlighted project constraints

	8/25	10/6	10/20	11/3	11/17
Controls **(Moustafa Khalil)**	Choose two-loop approach	Basic code for controller	Bench test motors/logging	Navigation added in	Validation of movement, balance, and safety
Power & Mechanical (Kaitlynn)	Chose battery and other power/mech needs	Finalize robot frame design and component layout	bench test converters & H bridge , assemble robot frame	Integrate motors with power system, test drive function	Validate power delivery, frame strength, and reliability
Microcontroller **(Gage Fox)**	Serial Communication	PWM, ADC, and GPIO handling	Bench Test motor control. Design PCB	Test sensor communication.	Validate comms., motor control, logic precision, power efficiency.
Sensors & Navigation **(Eric Chang)**	Define sensor algorithms and required components	Sensor interfaces (I^2C communication)	Test raw data. Design filtering and fusion algorithms	Test filtering algorithms with MATLAB	Validate sensors, filters, GUI. Compare with metrics